Adithya Ram S

Indian Institute of Information Technology Sri City
Phone: +91-6381426353 Email: adithya395672@gmail.com
GitHub: github.com/Adk-157 LinkedIn: linkedin.com/in/adithyarams Portfolio: adk-157.github.io/Portfolio-website

Education

- **B.Tech (Honours) in Electronics and Communication Engineering** 2023–2027
  Indian Institute of Information Technology Sri City CGPA: 8.43

Experience

- **Research Intern, Photonics & Biosensors Lab** Jun–Jul 2026
  Indian Institute of Technology Madras
  - Built a low-cost active chlorophyll fluorescence spectroscopy system as a proxy for solar-induced fluorescence-based plant health monitoring.
  - Ran a comparative multi-stressor study with quantified stress-response signatures and parallel control validation; authored a conference-style paper draft.
  - Proposed a Raspberry Pi NoIR camera-based low-cost spectrometer as an alternative to commercial instrumentation.
- **Software Intern, Hardware Platform Solutions** May–Jul 2025
  Celestica India Pvt. Ltd., Chennai
  - Extended an internal Angular-based test orchestration platform with end-to-end configurable test triggering, replacing a manual multi-step workflow; used Python and Angular.
  - Wrote shell-side scripts for server-triggered test execution and result logging, enabling automated regression runs without manual intervention.
- **Research Intern, Embedded Systems (Honours Project)** Jul–Aug 2025
  IIIT Sri City
  - Built Raspberry Pi firmware for synchronized GPS (UART) and IMU (I2C) acquisition using interrupt-assisted polling, achieving reliable multi-sensor logging under real mobility conditions.
  - Designed an end-to-end telemetry pipeline from embedded sensor sampling through wireless transmission to Firebase and Leaflet.js dashboards.

Projects and Research

- **FPGA Siren Detection Accelerator** 2026
  Team Dual Cores | BITS Hyderabad FPGA Hackathon – Poster Round Selectee
  - Deployed a quantized INT8 1D-CNN with FFT-based feature extraction on a Zynq-7020 (ZedBoard), using HLS optimization to achieve 31 μs inference latency for real-time, low-power siren detection.
- **4-bit SAR ADC** 2026
  Cadence Virtuoso
  - Designed and simulated a 4-bit Successive Approximation Register ADC at the transistor level in Cadence Virtuoso, covering comparator, DAC, and SAR logic sub-blocks.
- **IoT E-Cycle Telemetry System** 2025
  Provisional Patent Filed
  - Designed and deployed a Raspberry Pi telemetry system on a live e-cycle platform (GPS/IMU over UART/I2C, multi-hop LoRa/ESP-NOW transmission) with a complete hardware-to-dashboard pipeline streaming to Firebase; architecture novelty led to a provisional patent filing.
- **SKRAID: Skid Risk Assessment in Indian Driving Conditions** 2025
  Honours Research Project
  - Developing a Raspberry Pi-based embedded system with high-reliability IMU/GPS acquisition and Python preprocessing pipelines to produce labelled datasets for skid-risk classification under dynamic Indian driving conditions.

Technical Skills

Languages: Embedded C, C, C++, Python, Verilog HDL

Platforms: Raspberry Pi, ESP32, Arduino, PIC, 8051

Interfaces: UART, I2C, SPI, PWM, GPIO, LoRa, ESP-NOW

EDA and Tools: Vivado, Vitis, Cadence Virtuoso, Electric, LTSpice, Eagle PCB, Git

Domains: Embedded Systems, IoT Telemetry, Cyber-Physical Systems, VLSI Design, Mixed-Signal Circuits

Leadership and Certifications

- **Club Lead**, NexSync Smart Mobility Club, IIIT Sri City 2025–Present
- **Certifications:** Verilog HDL Masterclass (Udemy) | IoT and Autonomous Systems (IEEE ITSS) | ESP32 IoT Internship (COGTIS)